Code No: 136AW **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
CRYPTOGRAPHY AND NETWORK SECURITY
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is Authentication?	[2]
b)	What is cipher text?	[3]
c)	What is message authentication code?	[2]
d)	What elements of HTTPS are encrypted?	[3]
e)	What protocols comprise SSL?	[2]
f)	What are the principal services provided by PGP?	[3]
g)	What is DKIM?	[2]
h)	What are the different classes of Intruders?	[3]
i)	What is Honey pots?	[2]
j)	What is kerberos realm?	[3]

PART – B

(50 Marks)

2. List and briefly explain categories of security services and security mechanisms. [10]

OR

3. Explain steganography and explain different possible types of attacks. [10]

4. Explain Diffie – Hellman key exchange. [10]

OR

5. Explain RSA public key encryption algorithm. [10]

6. Explain X.509 certificates. Give its format. [10]

OR

7.a) Explain digital signatures.
b) Explain secure hash algorithm. [5+5]

JNTUH Used papers 2023

8.a) Explain Transport layer security.
b) Briefly define SSH protocols. [5+5]

OR

9.a) Explain about IEEE 802.11 wireless LAN.
b) Explain wireless security. [5+5]

10.a) Explain S/MIME.
b) Explain Internet key exchange. [5+5]

OR

11.a) Give IP security architecture.
b) Explain anti-replay service in ESP. [5+5]

---ooOoo---

JNTUH Used papers 2023